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Geology 1991;19;127-130
doi: 10.1130/0091-7613(1991)019<0127:MPSITU>2.3.CO;2

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Magnetic polarity stratigraphy in the uppermost Mississippian Mauch Chunk Formation, Pottsville, Pennsylvania

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ABSTRACT

We took 105 oriented cores from 340 m of the upper Mauch Chunk Formation at Pottsville, Pennsylvania, for a magnetostratigraphic study. The samples were subjected to progressive thermal demagnetization from 600 to 700 °C and analyzed by using principal-components analysis. A high-temperature component was isolated with a mean direction of declination = 351.7°, inclination = -28.4°, and α_{95} = 6.7°, and a corresponding paleopole at long 113.5°E, lat 33.7°N. Three normal and three reversed polarity magnetozones were defined, and it was found that the magnetic field was ~59% normal and 41% reversed in the upper Chesterian. Herein we attempt to correlate the section to a previous study in New Brunswick on the basis of the paleolatitude (pole position) and the reversal pattern, and we postulate a correlation of the bottom part of the Maringouin section with the top of the Mauch Chunk. Such a correlation would place the Mississippian/Pennsylvanian boundary within the Maringouin Formation. Earth's magnetic field was of normal polarity at the time of the Mississippian/Pennsylvanian boundary.

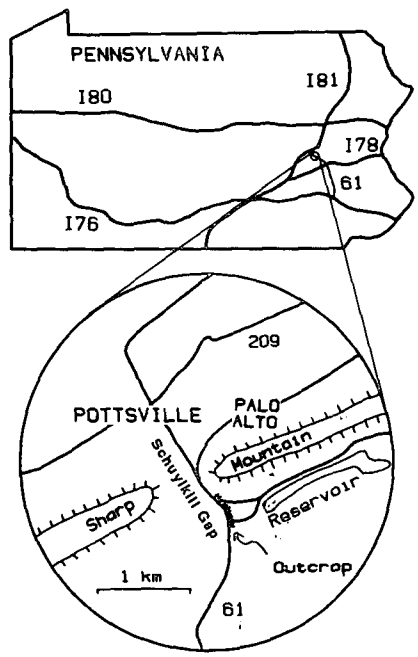


Figure 1. Location of sampled section at Pottsville, Pennsylvania.

INTRODUCTION

Earlier work in red sedimentary rocks of the Mauch Chunk Formation has identified two ancient components of magnetization (Kent and Opdyke, 1985). One component, with unblocking temperatures as high as 650 °C, was shown to have been acquired during folding in the Alleghenian orogeny (Kent and Opdyke, 1985; Kent, 1988). A prefolding characteristic component with unblocking temperatures normally over 650 °C was shown to carry both normal and reversed polarity at different sites.

Previous studies of the Mauch Chunk Formation have all been done in investigations related to tectonic problems (Irving and Opdyke, 1965; Knowles and Opdyke, 1968; Kent and Opdyke, 1985; Kent, 1988). This study is intended to test the feasibility of doing magnetic stratigraphy in the formation. The goal of this study is to construct a magnetic polarity stratigraphy from the high-temperature component and to attempt to correlate this stratigraphy to that developed in the Cumberland structural basin of Nova Scotia and New Brunswick (DiVenere and Opdyke, 1990, and unpublished).

GEOLOGIC SETTING

The Mauch Chunk Formation is a widespread unit of Late Mississippian age. It was divided informally into three members by Wood et al. (1969). The upper member intertongues with the overlying Pottsville Formation. The upper contact at Schuylkill Gap, placed by Levine and Slingerland (1987) at the last Mauch Chunk–type red bed, is the type locality where the Mississippian/Pennsylvanian boundary was originally defined (Wood et al., 1956; Moore et al., 1944). The beds, which are slightly overturned at Pottsville, were folded during the Alleghenian orogeny and are part of the Minersville synclinorium (Levine and Slingerland, 1987).

SAMPLING

We took 105 oriented cores in stratigraphic sequence from 340 m of the upper Mauch Chunk Formation beneath its upper contact with the Pottsville Formation. Sampling was done at Schuylkill Gap, southeast of Pottsville, Pennsylvania, on U.S. Highway 61 (Fig. 1). A sampling interval of 1 m was attempted but not achieved because of the presence of intraformational conglomerates, which were not sampled. The outcrop is continuous in a road cut except for a 22 m covered interval in the upper part of the section. The red mudstones, siltstones, and sandstones are slightly overturned at angles of up to 110° to the southeast.

Samples were obtained with a portable, gasoline-powered drill and were oriented with a Brunton compass. The 2.25-cm-diameter cylindrical samples were later cut into 1.25-cm-long specimens for measurement. A total of 22 strike and dip measurements were taken of bedding orientations through the section.

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TREATMENT

Demagnetization was performed in a Schonstedt model TSD-1 thermal demagnetizer, and the remanent magnetization was measured in a Superconducting Technologies two-axis cryogenic magnetometer at the University of Florida. Both are housed inside a magnetically shielded room.

The natural remanent magnetization (NRM) of the samples was measured, and the samples were then subjected to progressive thermal demagnetization beginning at 600 °C, because the low-temperature component was well documented by Kent and Opdyke (1985) and Kent (1988). Samples were usually heated at 15 °C steps to 630 °C, 10 °C steps to 660 °C, and 5 °C steps thereafter. Some samples were treated to a maximum temperature of 700 °C.

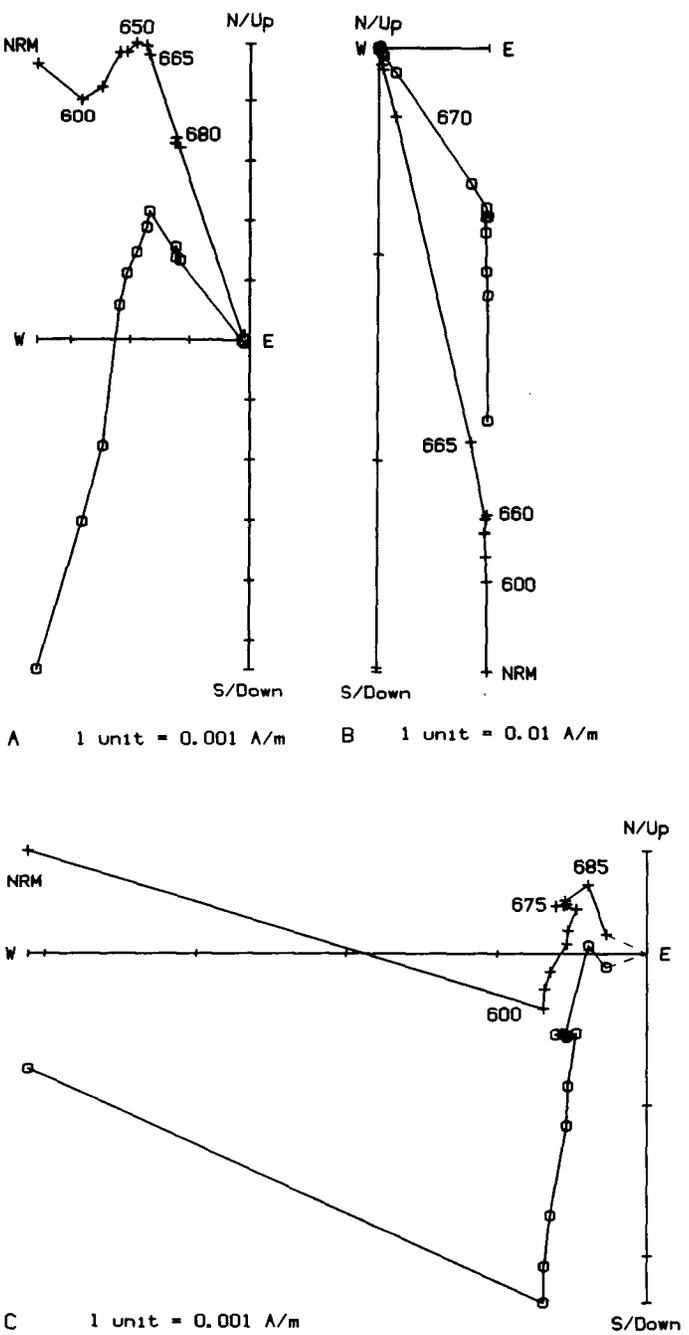


Figure 2. Orthogonal plots of demagnetization behavior corrected for bedding tilt showing (A) normal C component, (B) reversed C component, and (C) incompletely resolved normal C component. Crosses represent horizontal trace; open symbols represent vertical trace of magnetic vector. NRM is natural remanent magnetization.

RESULTS

Demagnetization Behavior

The typical measured intensity of the NRM was $\sim 10^{-2}$ A/m. Both normal and reversed polarities were observed before treatment. Upon progressive thermal demagnetization over 600 °C, a secondary (B) component, which is directed south with positive inclination, was seen in all samples (Fig. 2). This component was acquired during folding (Kent and Opdyke, 1985). After the removal of the B component, a high-temperature (C) component with dual polarities was isolated (Fig. 2, A and B). In some cases the high-temperature component did not yield a good straight line to the origin; however, in these cases the polarity was usually clear (Fig. 2C). Maximum observed temperatures at unblocking ranged from 670 to 690 °C.

Linear segments were chosen visually for principal-components analysis (Kirschvink, 1980). The C components with clear linear decays to the origin were chosen for Fisher (1953) statistics. The chosen C components are plotted on an equal-area projection in Figure 3. There is a degree of scatter, but two antipodal groups of points are seen, north and up, and south and down. A reversal test was performed by comparing the mean direction for the normal samples vs. the reversed samples (Table 1). The angular distance between the northerly and southerly groups (flipped to a common hemisphere) is 7.6°. The circles of confidence overlap at the 95%

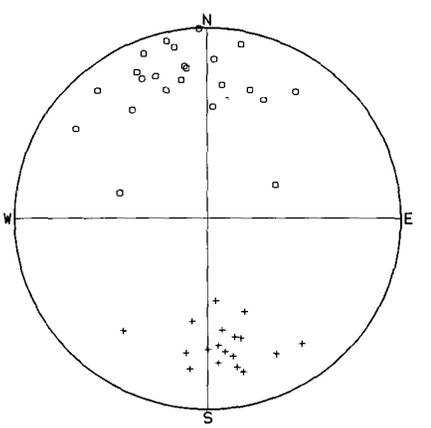


Figure 3. Directions of magnetization of C component plotted on stereographic projection after tilt correction. Crosses represent vectors that plot on lower hemisphere; circles plot on upper hemisphere. N = 41.

TABLE 1. MAUCH CHUNK FORMATION — CHARACTERISTIC COMPONENT

Polarity	D (°)	I (°)	α_{95} (°)	k	N	R
Normal	350.4	-25.1	10.8	8.8	23	20.509
Reversed	173.3	32.3	7.3	23.2	18	17.266
Combined	351.7	-28.4	6.7	12.1	41	37.692

Note: D is declination, I is inclination, α_{95} is 95% circle of confidence, k is Fisher's precision parameter, N is number of samples, R is length of resultant vector.

confidence level, and the directions are treated as a single population. Although no fold test was performed in the study, Kent and Opdyke (1985) showed that the C component is prefolding. The mean direction determined for the upper Mauch Chunk Formation is declination (D) = 351.7°, inclination (I) = -28.4°, and α_{95} = 6.7° for number of samples (N) = 41. The location of the mean paleomagnetic pole is long 113.5°E, lat 33.7°N, (δp = 4.0°, δm = 7.4°). The site paleolatitude is 15.1°S.

Magnetic Stratigraphy

All virtual geomagnetic poles (VGP) for samples with clear polarity (N = 101) were normalized by calculating the angular distance of each VGP from the formation mean pole. Plotting normalized VGP latitudes vs. stratigraphic position yields an internally consistent magnetic polarity stratigraphy with three normal (MMn1–MMn3) and three reversed (MMr1–MMr3) magnetozones (Fig. 4). The polarity is reversed at the bottom of the measured section and normal at the top of the Mauch Chunk Formation at its upper contact with the basal Pennsylvanian Pottsville Formation. The section is ~59% (199 m) normal and 41% (140 m) reversed polarity.

DISCUSSION

The study area is on the north limb of the Pennsylvania salient. Kent (1988) has shown that there has been oroclinal rotation around the salient since the acquisition of the C component in the Mauch Chunk Formation. The present result with a mean direction at D = 351.7°, I = -28.4°, and α_{95} = 6.7° is in good agreement with the direction determined by Kent and Opdyke (1985) from the north limb of the salient of D = 346.4°, I = -25.6°, and α_{95} = 9.7°. This is compared with a revised (after Kent and Opdyke, 1985) direction from the south limb of the salient of D = 323.1°, I = -30.4°, and α_{95} = 10° (Kent, 1988).

The present result (MCP) is compared with other Carboniferous paleopoles in Figure 5. The mean paleomagnetic pole at long 113.5°E, lat 33.7°N has a similar latitude to and is not significantly different from the early Namurian (Late Mississippian) Maringouin Formation (Ma in Fig. 5) of New Brunswick (DiVenere and Opdyke, 1990), which has a paleopole at long 121.3°E, lat 32.4°N. The paleopole is closest to the mean pole previously determined for the north limb of the Pennsylvania salient from the Mauch Chunk Formation (MCN) by Kent and Opdyke (1985). These poles are compared to the mean poles from the south limb of the Pennsylvania salient (MCS) (Kent, 1988), the Visean Deer Lake Formation (DL) (Irving and Strong, 1984), the middle Mississippian Pride Mountain Formation (PM) (McCabe et al., 1989), and the mid- to late Namurian Shepody Formation (Sh) (DiVenere and Opdyke, 1990). The 200, 240, 250, and 300 Ma mean poles from Irving and Irving (1982) are included to show the approximate trend of the late Paleozoic apparent polar wander (APW) path for North America.

It is not possible to determine, from the Mauch Chunk results alone, whether one or both limbs of the Pennsylvania salient are rotated; however, the results from the north limb are closest to those from the Canadian Maritimes. This may indicate that the north limb of the Pennsylvania salient is either unrotated or is rotated to a lesser degree than the south limb.

North America seems to have changed paleolatitude rapidly, ~5–7 cm/yr northward during middle Carboniferous time (Opdyke and DiVenere, unpublished). Therefore, the paleolatitude and associated paleopole position might be used with magnetostratigraphy to correlate geo-

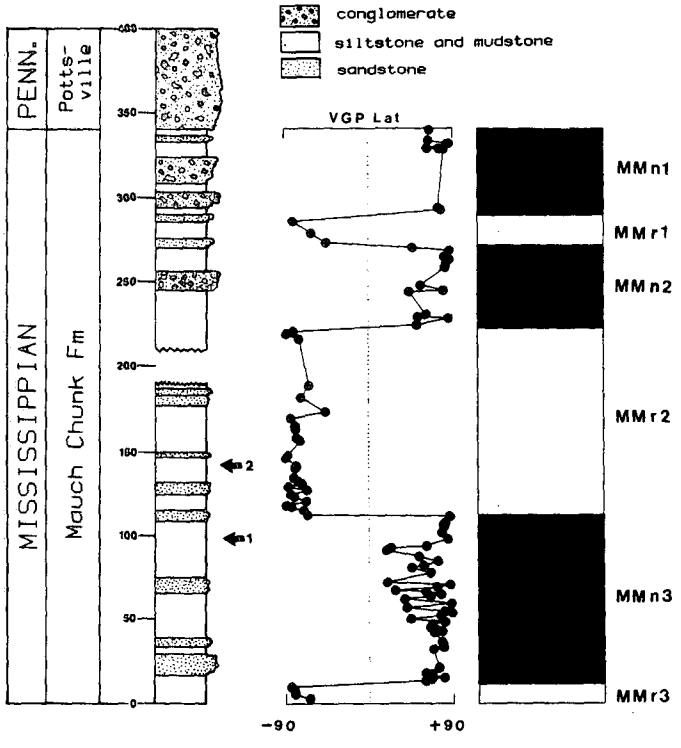


Figure 4. Magnetic stratigraphy of Mauch Chunk Formation at Pottsville, Pennsylvania, with virtual geomagnetic pole latitudes (normalized to section mean) plotted against stratigraphic position (N = 101). Generalized stratigraphy is separated into sandstones, Mauch Chunk–type red beds, and Pottsville-type conglomerates. Normal magnetozones MMn1 through MMn3 and reversed magnetozones MMr1 through MMr3 are labeled. 1—telephone pole; 2—steel grate.

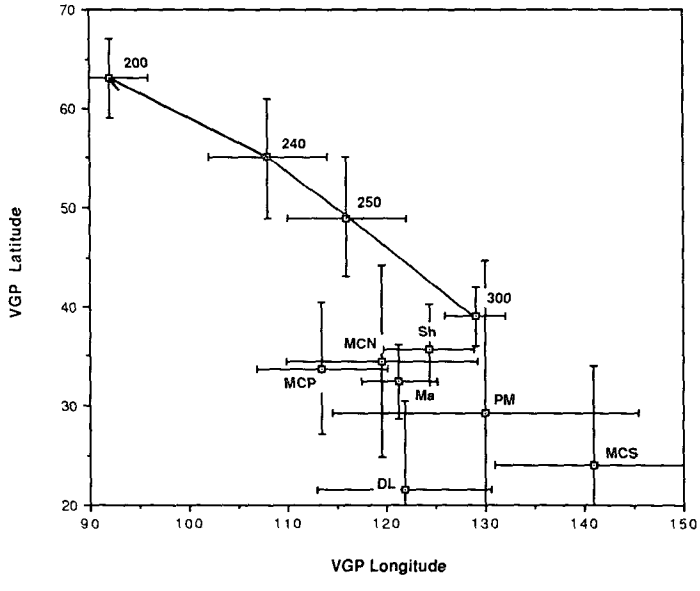


Figure 5. Paleopole positions from Carboniferous studies. MCP = this study, MCN = Mauch Chunk north (Kent and Opdyke, 1985), MCS = Mauch Chunk south (Kent, 1988), DL = Deer Lake (Irving and Strong, 1984), PM = Pride Mountain (McCabe et al., 1989), Ma = Maringouin, and Sh = Shepody (DiVenere and Opdyke, 1990); 200, 240, 250, and 300 Ma poles from Irving and Irving (1982). Error bars are α_{95}.

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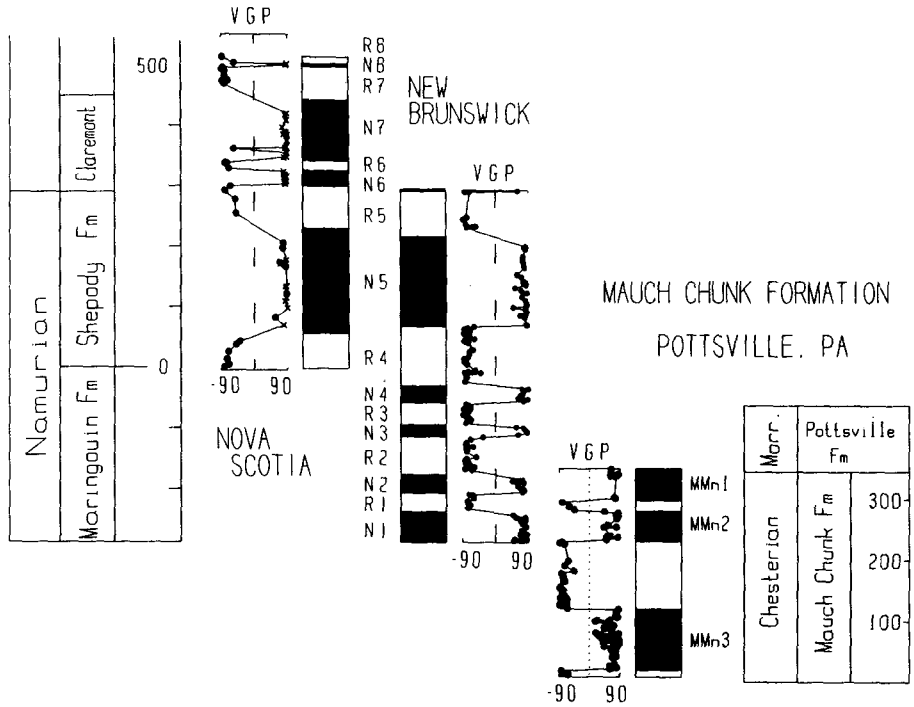


Figure 6. Tentative correlation between Mauch Chunk Formation at Pottsville, Pennsylvania, and Maringouin Formation from Cumberland structural basin of New Brunswick and Nova Scotia (DiVenere and Opdyke, 1990, and unpublished). In this representation, MMn1 of Mauch Chunk Formation is correlated with N2 of Maringouin Formation. Alternatively, MMn1 could be correlated with N4.

graphically distant stratigraphic sections. The Mauch Chunk Formation paleopoles from the north limb of the Pennsylvania salient are most similar to that derived from the Maringouin Formation. The Mauch Chunk and Maringouin Formations have previously been correlated to the uppermost Mississippian (Moore et al., 1944). We suggest that the upper two normal zones described in the Mauch Chunk Formation (MMn1, MMn2) may correlate to N1 and N2 in the Maringouin Formation (Fig. 6). An alternate assignment could be N3 and N4. However, both sections must be extended downward to allow a positive correlation.

Eastern North America (New Brunswick) was still at 8°–9°S paleolatitude (DiVenere and Opdyke, unpublished) in earliest Pennsylvanian time, according to the above tentative correlation, yet reached the equator by the Late Pennsylvanian or Early Permian. This would indicate ca. 0.2°/m.y. or 3 cm/yr of average northward drift during the Pennsylvanian.

CONCLUSIONS

The section mean direction yields a paleomagnetic pole at long 113.5°E, lat 33.7°N, which is in good agreement with that obtained by Kent and Opdyke (1985) from the Mauch Chunk Formation on the north limb of the Pennsylvania salient. A magnetic polarity stratigraphy was obtained with three normal and three reversed magnetozones and ~59% normal and 41% reversed polarity. The upper Mauch Chunk Formation is tentatively correlated with the Maringouin Formation of New Brunswick on the basis of a comparison of paleomagnetic pole latitudes and magnetostratigraphy.

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Manuscript received May 3, 1990
Revised manuscript received September 4, 1990
Manuscript accepted September 12, 1990